THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS · HARD LEVEL CHAPTER TEST SET 3

Chapter 2: **Linear Polynomials** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. If $x = 3$ is the zero of $kx - 33$, then $k =$ 1

(a) 13 **(b)** 10 **(c)** 12 **(d)** 11

2. A linear polynomial whose zero is -3 and which takes value 6 at $x = 0$ is: 1

(a) $2x - 6$ **(b)** $x + 6$ **(c)** $3x + 6$ **(d)** $2x + 6$

3. For the pattern with nth term $9n - 4$, the sum of the 1st and 8th terms is: 1

(a) 73 **(b)** 74 **(c)** 72 **(d)** 75

4. For which value of n is $14n - 9 = 257$? 1

(a) 19 **(b)** 18 **(c)** 20 **(d)** 21

5. The area (in square units) of the triangle formed by the line $y = 10x + 30$ and the two axes is: 1

(a) 46 **(b)** 45 **(c)** 47 **(d)** 44

6. The x-coordinate of the point where $y = 8x + 3$ and $y = 4x + 27$ meet is: 1

(a) 5 **(b)** 6 **(c)** 7 **(d)** 8

7. If $p(x) = 2x - 5$, then $p(p(6)) =$ 1

(a) 9 **(b)** 11 **(c)** 8 **(d)** 10

8. A plan costs ₹60 plus ₹30 per unit; another costs ₹360 plus ₹10 per unit. They cost the same for ___ units. 1

(a) 15 **(b)** 16 **(c)** 17 **(d)** 14

9. **Assertion (A):** The nth term of the sequence $5, 8, 11, 14, \ldots$ is $3n + 2$. 1

Reason (R): The terms increase by a constant amount 3, so the nth term is a linear polynomial in n with coefficient 3.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

10. **Assertion (A):** The value of the polynomial $p(x) = 5x - 2$ at $x = 2$ is 8. 1
Reason (R): A linear polynomial has exactly one zero.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Find the area of the triangle formed with the axes by the line $y = 6x + 12$. 2

12. Plan A costs ₹500 plus ₹55 per unit; Plan B costs ₹740 plus ₹35 per unit. For how many units do both cost the same? 2

13. If $p(x) = 5x - 2$, find $p(p(x))$ and its value at $x = 3$. 2

14. Find the area of the triangle formed with the axes by the line $y = 10x + 50$. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Find the point of intersection of $y = 9x + 4$ and $y = 4x + 34$. 3

16. A linear polynomial has $p(1) = 21$ and $p(3) = 35$. Find $p(x)$, its zero and $p(2)$. 3

17. The nth term of a pattern is $18n + 1$. Find the number of terms that are less than 145 and the sum of the first two terms. 3

OR

Find the point of intersection of $y = 8x - 5$ and $y = 3x + 15$. 3

18. A linear polynomial has $p(1) = -8$ and $p(3) = 8$. Find $p(x)$, its zero and $p(5)$. 3

OR

The nth term of a pattern is $13n + 6$. Find the number of terms that are less than 149 and the sum of the first two terms. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** A water tank in a school holds 120 cm of water. It is being emptied by a pump so that the height of water after t minutes is $h(t) = 120 - 4t$ centimetres. 5
(i) What is the height at the start?
(ii) What is the height after 10 minutes?
(iii) Find the zero of $h(t)$.
(iv) Find the slope and intercepts of the graph of $h = 120 - 4t$ and interpret the slope.

20. Two taxi services: A charges ₹35 plus ₹13 per km; B charges ₹95 plus ₹8 per km. Find the distance for which both charge equally, the common fare, and which is cheaper for 10 km. 5

OR

A line passes through $(2, 35)$ and $(8, 77)$. Find its slope, equation, intercepts and the area of the triangle it forms with the axes. 5

END OF QUESTION PAPER — **Total: 40 Marks**